COUNTERFEIT INTEGRATED CERCUITE (IC) DETECTION USING IMAGE PROCESSING

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This project report is hereby approved as a creditable study of an engineering subject carried out and presented in a manner satisfactory to warrant its acceptance as a pre-requisite to the degree for which it has been submitted. It is to be understood that by this approval, the undersigned do not necessarily endorse or approve any statement made, opinion expressed and conclusion drawn therein but approve the project report only for the purpose for which it has been submitted.

COMMITTEE ON FINAL EXAMINATION FOR EVALUATION OF PROJECT REPORT

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ABSTRACT

The widespread penetration of counterfeit integrated circuits (ICs) is not only a major threat to the electronic goods supply chain, but also constitute a great threat to national security. With the IC market becoming cheap, we see a boom in sales and production. The customers of the market vary from low scale research labs to multinational giants producing daily electronics. Apart from sophisticated technologies, even the daily electronic goods get hampered and there is a mass scale production of these low scale and less durable goods. This is why the detection and consequent elimination of such counterfeit ICs is of utmost importance. But such a method of detection should have qualities like high usability, high flexibility, intuitive and efficient with respect to computation time and complexity.

In this report, we first briefly state the current market scenario and challenges pertaining to limiting the production of counterfeit ICs. We describe in detail different kinds of such ICs produced and have attached surveys conducted by IHS (Information Handling Service) regarding the growing problem of counterfeit products. We also briefly mention the various methods already available for distinction and detection of such ICs and how it is different from the methods we implement in this project. We also describe briefly how to distinguish between original and counterfeits based on the properties of texture, quality and structure and how other such defects can be detected from external observation by capturing its image and comparison with the original products.

In this project, we try to implement such a detection technique using image processing and computer vision techniques which although is flexible, usable and intuitive come with its own set of challenges. Image processing based counterfeit IC design techniques are promising, but currently often suffer from high computational complexity and requirement of expensive image acquisition infrastructure. We describe two techniques based on image segmentation and Deep Learning to automate the process of counterfeit IC detection. The first method employs areal mask by segmenting the image into appropriate regions of the IC body to detect counterfeit ICs. The second method includes identification of counterfeit ICs by using Deep Neural Network to analysis the image shape and property. The first method is of lower computational complexity compared to the segmentation method, but the second method is a high-level image processing method. Our experimental results show that these methods have high detection accuracy and we expect that it will be able to detect all kind of counterfeiters for any IC and this process is still in development progress. In this paper to distinguish between normal IC and counterfeit ICs we mainly focused on scratches, defects in the IC pin.

Keywords—Counterfeit IC detection, image processing, mask based segmentation, deep neural network.

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"A mediocre person tells. A good person explains. A superior person demonstrates. A great person inspires others to see for themselves"’

-Harvey Mackay

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1. INTRODUCTION

A worldwide epidemic of counterfeit electronic components is flooding the market and affecting the supply chains of all industries. It is estimated that the financial loss due to counterfeit components is over $100 billion per year. Most of the effort today has not been placed on preventing counterfeiting but rather on screening components to identify and remove counterfeits before they are used in a finished product.

In 2001, the International Chamber of Commerce (ICC) estimated that 5–7% of world trade was in counterfeit goods and that the counterfeit market was worth $350 billion. Based on 2008 data, it was estimated that counterfeit and pirated products could account for as much as US$650 billion per year globally. Due to the rapid increase in counterfeiting and piracy, it was predicted

OECD category	Estimate (2008 data)	Estimate (2015)
Internationally traded counterfeit and pirated products	$285–360 billion	$960 billion
Domestically produced and consumed counterfeit and pirated products	$140–215 billion	$570 billion
Digitally pirated products	$30–75 billion	$240 billion
Total	$455–650 billion	$1.77 trillion

that the global market for counterfeit goodies likely to be more than double to US $1.7 trillion by 2015. Following table shows the breakdown of these estimates.

In the US, the Customs and Border Protection (CBP) agency is responsible for the seizure of counterfeit and pirated products that are imported and could possibly infringe US patents, trademarks, copyrights and other forms of intellectual property. Recent data from the CBP shows that the number of Intellectual Property Rights (IPR) seizures in 2013 increased by nearly 7% from 2012.The People's Republic of China remains the primary source for producing counterfeit and pirated goods. It represents approximately 68% ($1.1 billion) of all IPR seizures based on the manufacturer's suggested retail price (MSRP) in 2013. The counterfeit trade in computers and accessories went from $34,710,624 MSRP to $47,731,513 MSRP. Similar growth can also be seen in the counterfeit trade of consumer electronics and parts, which, as of FY 2013, was ranked behind only handbags, wallets, watches, and jewelry in its presence in the counterfeit market. More generally, total seizures increased 38%, from $1,262,202,478 to $1,743,515,581 between FY 2012 and FY 2013.

While the scope of the counterfeit trade is difficult to assess due to its largely clandestine nature, there have been many recent—and troubling—incidents when the counterfeit trade and its victims have faced before the public eye. In 2006, the World Health Organization (WHO) estimated that between 10% and 30% of the medications circulating in developing countries

were counterfeit. This can literally be a life-and-death problem for consumers. For instance, cough syrup containing ethylene glycol was identified as being responsible for the deaths of hundreds of people in Panama and the Dominican Republic in the span of less than a week. From highly visible status symbols to dangerously invisible parts of the systems we rely on for our health and safety, counterfeiting is a far-reaching problem that requires a coordinated response. In recent years, the "business" of counterfeiting has changed from the piecemeal production of inferior goods in small, clandestine workshops to the coordinated and sophisticated production of goods that are becoming harder and harder to differentiate from "the real thing." A case in point is that of Japanese electronics giant NEC: when NEC found out that pirated keyboards, CDs, and DVDs bearing the company logo were being circulated on the black market in Beijing and Hong Kong, they thought that they had a run-of-the-mill pirating threat on their hands. What initially seemed like a routine intellectual property (IP) concern, however, turned out to be an ambitious counterfeiting campaign. Instead of just faking items sold by NEC, the counterfeiters were faking the entire company. The NEC counterfeiters established, what was effectively a parallel brand where they not only copied NEC's product line but also developed their own range of consumer electronic products. The counterfeiters ran this parallel business like any normal business: they had business cards printed, commissioned new product research and development, signed production and supply orders, and issued official-looking warranty and service documents. As president of International Risk (a Hong Kong based company hired by NEC to investigate the piracy) Steve Vickers observed, the NEC case shows how drastically piracy is evolving. It has, in Vickers' words, "gone from often shoddy copying of brands to highly coordinated operations of production and marketing." In a separate case of counterfeiting, the CBP and ICE also made more than 1,300 seizures of semiconductor devices that were falsely marked as military/aerospace grade and were also affixed with trademarks of reputed semiconductor companies. As of 2010, this initiative, which is a collaborative effort between the Federal Bureau of Investigation (FBI), Immigration and Customs Enforcement (ICE), Customs and Border Protection (CBP) and international agencies such as China's Ministry of Public Security (MPS), has led to 30 felony convictions and the seizure of nearly $143 million worth of counterfeit network hardware. In a majority of the incidents, a bulk of the seized counterfeit hardware came in the form of Cisco networking equipment, which was intended for use in US military computer networks and IT infrastructures for firewall protection and secure communication.

1.1 MOTIVATION

The problem with counterfeit ICs is that they are very hard to detect and mass produced goods easily enter the industry, labs and research without any significant monitoring. As a result we end up with electronics which are less durable and more prone to malfunctioning. The difficulty comes with monitoring such Ics as while buying, they are sold in batches and so it is physically impossible to check and match each of them individually. What my project aims is to find an efficient and fast solution to this. Using image processing techniques to develop a tool which can

detect any counterfeit IC by its image. My aim is also to make this method user friendly, flexible and intuitive. A tool such as that would help to cut down the illegal markets and would help to minimize the risk of our electronic goods being contaminated by those products.

1.2 BACKGROUND

In Babylon and ancient Egypt, for instance, priests hoped to increase their own legitimacy and proceeds by placing inscriptions from earlier civilizations on their monuments, thereby creating a "false authority". Both trademarks and the counterfeiting of those trademarks stretch back into ancient times, with Pliny the Elder describing the popularity of counterfeit coins as collector's items within Roman society. The use of counterfeit coinage for illicit trade such as smuggling and certain types of foreign trade was the norm in sixteenth- and seventeenth-century Genoa. One of the most famous examples of coinage counterfeiting took place in Renaissance France when supporters of the Pope directed parallel minting in order to undermine the authority of France's Protestant king.

While the counterfeiting of money is—even now—perhaps the best-known form of counterfeit activity, evidence suggests that product counterfeiting may have been an even older form of illicit trade. During the first three centuries of the Roman Empire, oil lamps were marked with the FORTIS brand name, which scholars speculate was widely copied, given the widespread use of this stamp on many artifacts. In recent years, the scope and magnitude of counterfeiting has exploded in size andvolume.InJuly2007,thePublicSecurityBureauofChinaandtheFBIfound$2 billion worth of counterfeit Microsoft software, including 19 versions of products in 11 languages, in a warehouse in southern China where workers assembled disks, authenticated materials and manuals, and prepared them for shipping, in what was probably the biggest counterfeit software bust in history. These counterfeit products turned up in 36 countries across six continents.

Thus, from the days of the ancient Rome to the rampant counterfeiting of products in the twenty-first century, counterfeiting has constantly evolved along with the advances in manufacturing and trade and is indeed an issue that needs to be addressed.

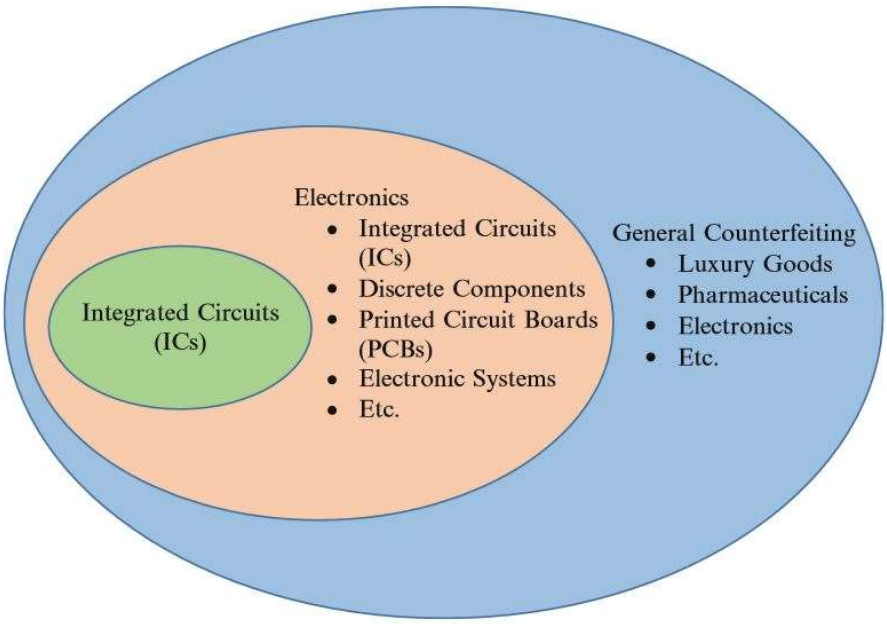
1.3 Counterfeit Business and Operational Network Raider

Though counterfeiting is not a problem specific to our historical moment, the stakes of counterfeiting are especially high in an age where one faulty link in the chain of systems that protect and enmesh us can damage the lives of countless people around the world. The high stakes of modern counterfeiting are particularly evident in the case of "Operation Network Raider", a domestic and global initiative that aims to control the illegal distribution of counterfeit network hardware.

1.4 Counterfeit Electronics

A wide variety of products we use in our day-to-day lives are subjected to counterfeiting. The following figure shows a broader category of counterfeit products, consisting of such things as luxury goods, pharmaceutical products, and electronics. Among them, electronics remain one of

the major counterfeit products, where components such as integrated circuits (ICs), discrete components, and printed circuit boards (PCBs) are frequently counterfeited.

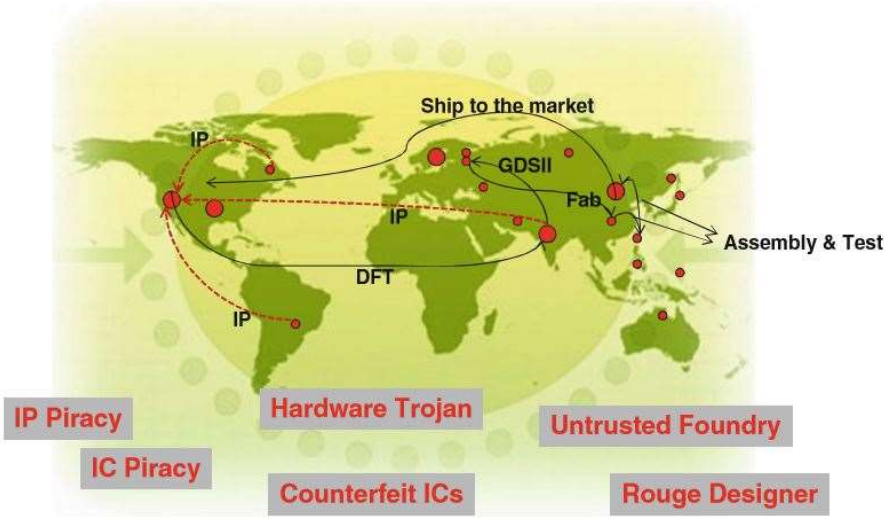


2. COUNTERFEIT INTEGRTED CIRCUITS

Counterfeit integrated circuits (ICs) pose a major concern to the industry and government as they potentially impact the security and reliability of a wide variety of electronic systems. in point of IC counterfeiting is Peter Picone, a 41-year-old man from Methuen, Massachusetts, who pled guilty in 2014 to importing thousands of counterfeit integrated circuits (ICs) from China and Hong Kong in order to resell them to US customers. The ice detection system on an aircraft is a critical module that alerts pilots about the presence of ice on an aircraft's control surface, a potentially life-threatening situation. The module was found with an FPGA unit that had literally fallen out of its sockets and was found inside the module. On further inspection, it was found that the Xilinx FPGA component, which was badly used, worn-out, and was a discontinued model, was sold to BAE Systems, the P-8 aircraft component contractor, by Tandex Test Labs in California. Tandex had bought the component through an independent distributor that had acquired the part through a manufacturer in Shenzhen, China. Further investigation revealed that the module was reworked and had found its way through the supply chain to the P-8A aircraft. Unquestionably, the presence of such inferior and untested ICs in US Navy nuclear submarines could have catastrophic and far-reaching consequences. Because counterfeit ICs are vulnerable to unpredictable failures, they can lead to property damage, costly repairs, and bodily injury—even in the most benign of circumstances. However, when counterfeit ICs are used in systems of national importance, they also raise several national security concerns. Counterfeit ICs' histories are unknown, so it becomes unclear who has had access to them and how they have been altered. Such devices can be changed so that they contain malicious code or hidden "backdoors" that can disable systems, intercept communications, and intrude into computer networks. All of these issues imply massive consequences when placed in the context of US national security.

Due to the complex nature of the component supply chain, it is impossible to estimate the size of the actual semiconductor counterfeit market. Most estimates are based on data derived from the number of seized or detected components. Since a large portion of counterfeit semiconductor components could be circulating

in the market, exact estimates are difficult to make. Following figure reflects the trust and security



Trust and security issues due to globalization

issues evident during the design and fabrication of electronic components. The designers use different IPs, collected from all across the globe, in their designs. It is extremely challenging and even impossible in some instances to validate their authenticity. The design-for-testability (DFT) for these integrated circuits is often inserted by third parties located in different places. Untrusted foundries and assemblies can also be capable of selling extra components outside of the number they were contracted to manufacture. Thus, this complex supply chain leads to an illicit market willing to undercut competition with counterfeit parts.

With counterfeit incidents on the rise, it is increasingly important to underst and what ICs are most likely counterfeit and what industries are impacted the most. Following table shows the five most commonly counterfeited components that represent $169 billion in potential annual risk for the global electronics supply chain. The components are as follows: analog ICs, microprocessor ICs, memory ICs, programmable logic ICs, and transistors.

Rank	Commodity type	% of reported incidents
#1	Analog IC	25.2
#2	Microprocessor IC	13.4
#3	Memory IC	13.1
#4	Programmable logic IC	8.3
#5	Transistor	7.6
#6	Others	32.4

Source: IHS parts management 2012

Following table shows the industries where these top five components are used. They include computing, consumer electronics, wireless and wired communications, automotive and industrial

sectors. Automotive and industrial sectors involve critical systems and, thus, the appearance of unreliable counterfeit components in these applications is quite alarming.

Part type	Industrial (%)	Automotive (%)	Consumer (%)	Wireless (%)	Wired (%)	Compute (%)	Other (%)
Analog IC	14	17	21	29	6	14	0
Microprocessor IC	4	1	4	2	3	85	0
Memory IC	3	2	13	26	2	53	1
Programmable logic IC	30	3	14	18	25	11	0
Transistor	22	12	25	8	10	22	0

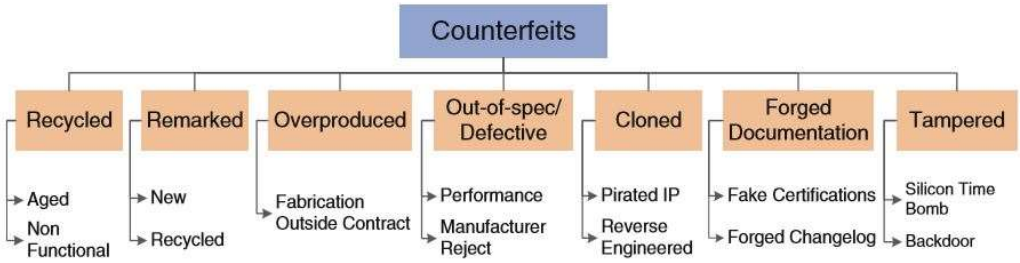
Percentage of market revenue for most commonly counterfeited product types by application market in 2011 (percentage share of revenue in millions of U.S. dollars)

2.1. Taxonomy of Counterfeit ICs

The US Department of Commerce define a counterfeit component as one that

- is an unauthorized copy
- Does not conform to original OCM design, model, and/ or performance standards
- is not produced by the OCM or is produced by unauthorized contractors;
- is an off-specification, defective, or used OCM product sold as “new” or working; or
- has incorrect or false markings and/or documentation.

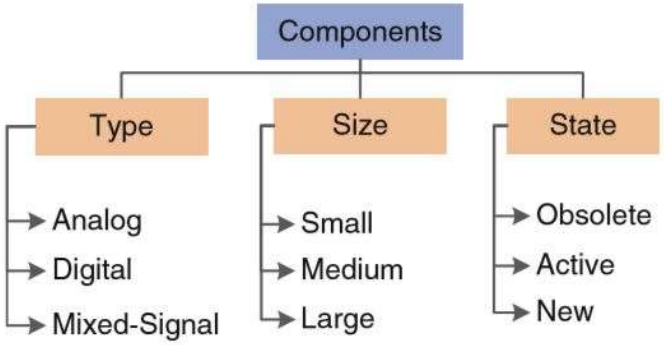
The above definition does not include all possible scenarios where an entity in the component supply chain source electronic components that are authentic and certified by the OCMs. we have expanded on the above definition of counterfeiting and developed a more comprehensive taxonomy of counterfeit types. The following figure shows this novel taxonomy of counterfeit types.



2.2 DETECTION AND AVOIDANCE OF COUNTERFEIT ICs

The detection and avoidance of counterfeit parts is a multifaceted problem, and we believe that it is still in its infancy. Currently, efforts at detection have mainly focused on identifying counterfeit parts that are already circulating in the electronic component supply chain by performing a sequence of “post-counterfeit” detection methods. On the other hand, avoidance

measures (also termed as design-for-anti-counterfeit (DFAC) measures) can help to prevent these counterfeit parts from entering into supply chain in the first place.



Taxonomy of component types

These components would only be available through OCM authorized or independent distributors of electronic components. Active components are still being manufactured by OCMs, but their designs cannot be changed because of

(1) The extra cost of developing new masks and

(2) Performance and reliability concerns. Thus, both obsolete and active components present very little opportunities for integration of anti-counterfeiting measures.

New components are very flexible in implementing avoidance measures as they are in the design phase where the OCM can still modify masks and validate performance.

2.2.1 Current Status of Detection

A few standards available today to guide users in detecting counterfeit parts. The group responsible for many of these standards is the G-19 Counterfeit Electronic Parts Committee, set forth by SAE International. Their standards target three different sectors of the industry: distributors, users, and test service providers (i.e., test laboratories). These standards are as follows:

1. AS5553—Counterfeit Electronic Parts; Avoidance, Detection, Mitigation, and disposition;

2. ARP6178—Fraudulent/Counterfeit Electronic Parts; Tool for Risk Assessment of Distributors;

2.4 Detection and Avoidance of Counterfeit ICs 31

3. AS6081—Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition—Distributors Counterfeit Electronic Parts; Avoidance Protocol, Distributors (intended for independent distributors and brokers);

4. AS6496—Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition—Authorized/Franchised Distribution; and

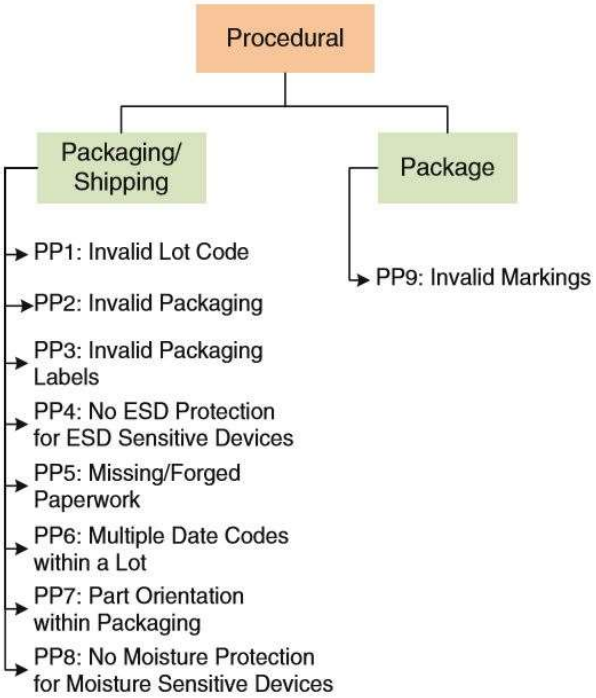
5. AS6171—Test Methods Standard; Counterfeit Electronic Parts.

Following figure shows the current technologies available for the avoidance of counterfeit parts in the component supply chain.

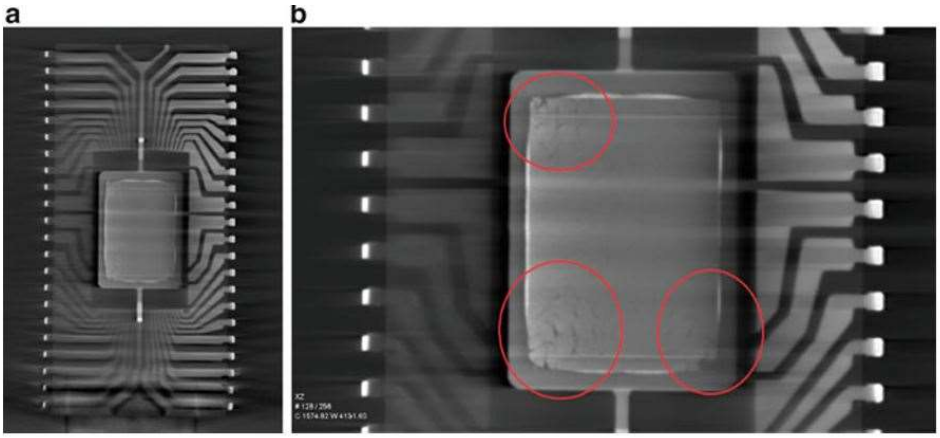
		Recycled	Remarked	Overproduced	Out-of-Spec/ Defective	Cloned
	Digital & Small	RO-CDIR, F-CDIR, DNA, NR	RO-CDIR, F-CDIR, DNA, NR, ECID			DNA, NR
	Transistors, Diodes, and Passive Parts	DNA, NR				
Digital & Large	Programmable Logic ICs Memory ICs Microprocessor ICs	All CDIRs, DNA, NR	All CDIRs, PUF, HM, SST, ECID, DNA,NR	HM, SST	SST	PUF, HM, SST, DNA, NR
	Analog & Mixed Signal ICs	F-CDIR, DNA, NR	F-CDIR, DNA, NR			DNA, NR

2.3 Defects:

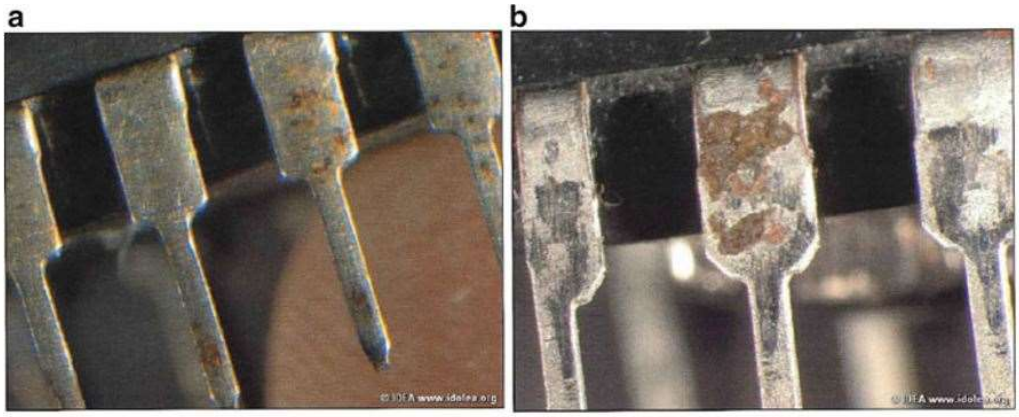
Awidevarietyoftestmethodsarecurrentlyavailablefordetectingcounterfeitparts. The goal of some methods is to identify “defects” presenting a part or a batch of parts under investigation. Counterfeit defects are those anomalies and changes that are not typically found in authentic parts.



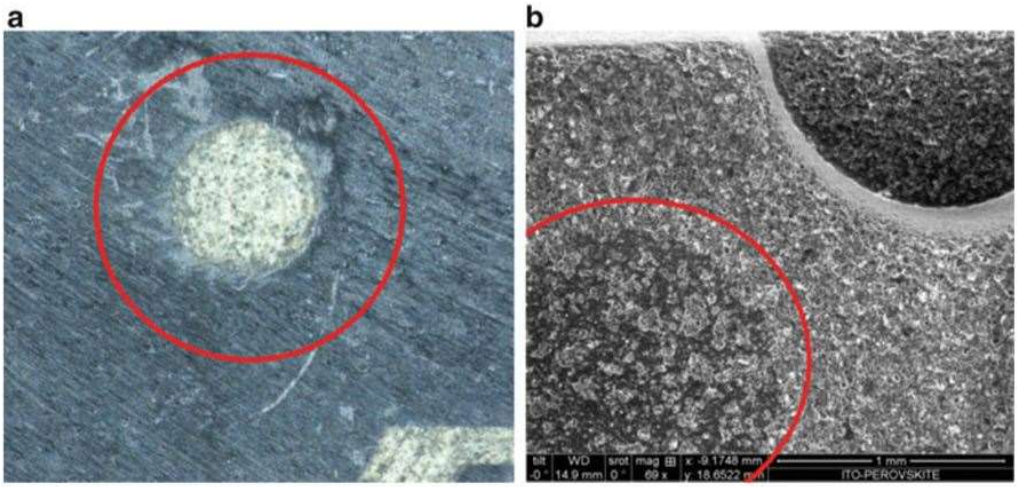
The various defects of which signifies counterfeit IC are shown in the next page.



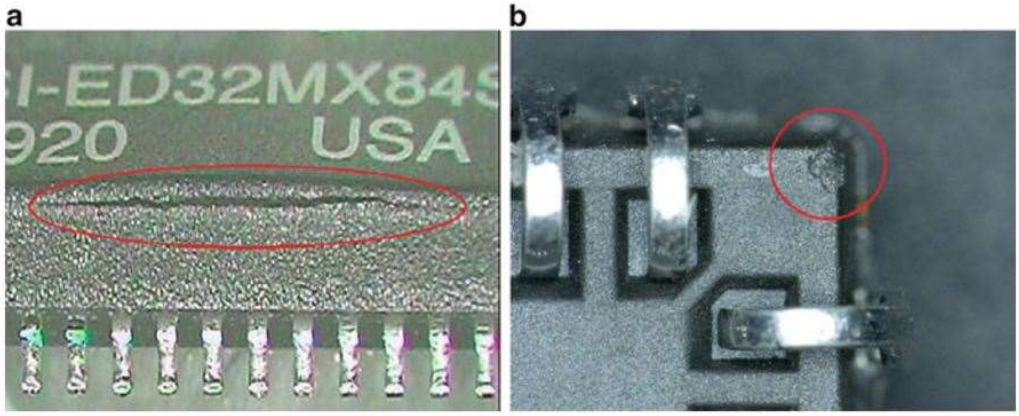
Damaged die (a) X-Ray image of an Intel flash memory (b) Damaged corner encircled in red



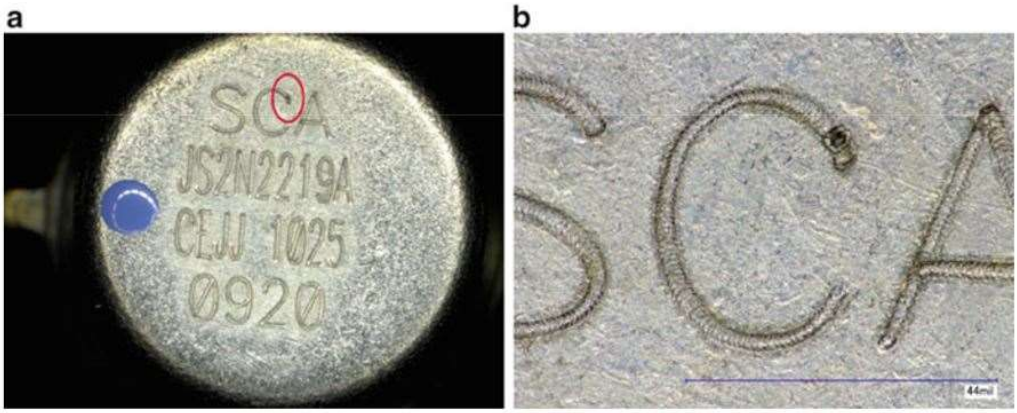
Oxidation on the leads of an IC [5](a) Lead contamination by oxidation (b) Oxidation on shoulder of lead



Sanded and partially filled dimples. (a) Sanded dimple, (b) filled dimple



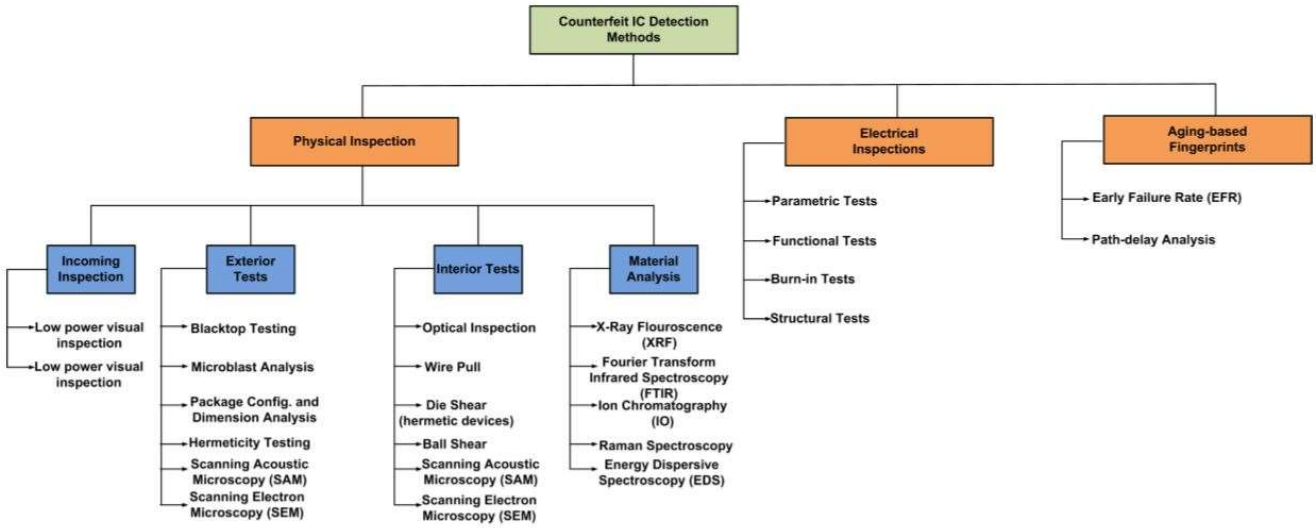
Package damage. (a) External crack, (b) chip out at the corner



Burned markings [Source: Honeywell Inc.] (a) Burned mark encircled in red (b) Magnified location of the burned mark

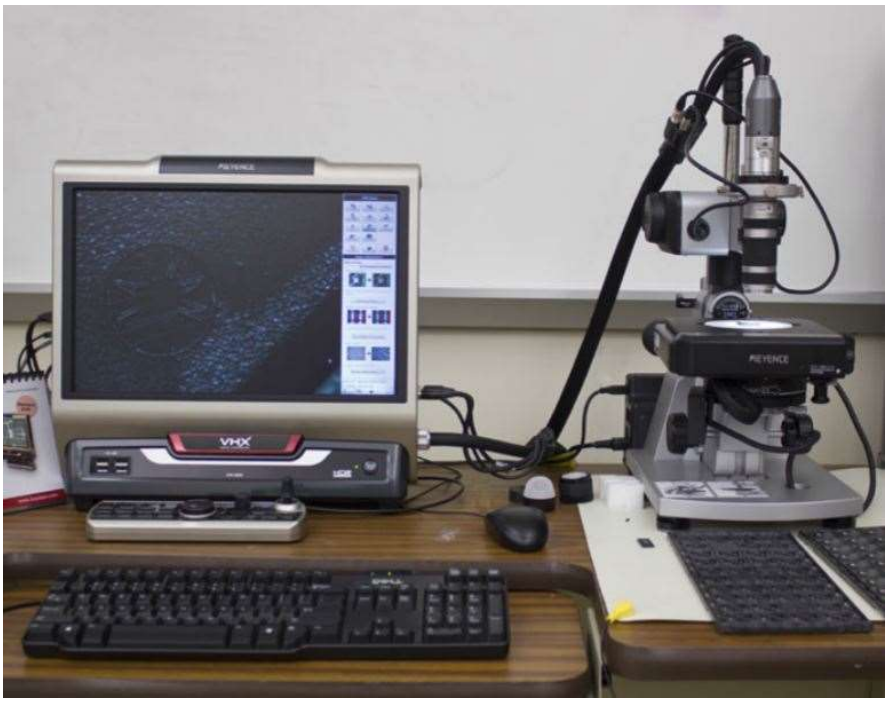
3. INVENTED TESTS FOR COUNTERFEIT IC DETECTION

The following figure shows a taxonomy of counterfeit detection methods.



3.1 DETAILED EVI

Detailed EVI is performed on the sampled components with a microscope with 10X–40X magnification. Higher magnification (up to 100X) may be necessary to spot some of the finer defects. for detailed visual inspection. The setup consists of a Keyence VHX-2000 digital microscope which is a versatile imaging and measurement system with 0.1X– 5000X magnification.



An inspection set up for detailed EVI (CHASE Center, University of Connecticut)

3.2 ZEISIS XRADIA 510 VERSA

Zeiss Versa 510 (which has a maximum power of 160kV for its source), at the University of Connecticut which has been used to acquire the structural information of different ICs. The ZEISS Xradia 510 Versa 3D X-ray Microscopes (XRM) provides a

Zeiss Xradia 510 Versa: X-ray imaging system (CHASE Center, University of Connecticut)

Unique RaaD (resolution at a distance) capability that breaks the one micron resolution barrier for samples from mm to cm. The system is used to capture two dimensional (2D) projections and X-Ray computed tomography to observe the detailed three dimensional (3D) internal structure of an IC.

3.3 CURVE TESTING

Curve tracing is gaining popularity in the detection of counterfeit components as it tests ICs non-destructively and without requiring extensive details of the ICs under test. The traces follow the Ohm's Law of V = I * Z, where Z is the impedance between the pins of an IC. shows a typical curve tracer by National Instruments. This system can perform sweeps across different IC pins with a range of -20 to +20V.

A typical curve tracer by National Instruments

Pin No	V Spot 1 (V)	I Spot 1 (mA)	V Spot 2 (V)	I Spot 2 (mA)	Result
13	−0.094	−0.605	0.086	0.592	Spot Fail
28	−0.094	−0.605	0.086	0.592	Spot Fail
40	−0.090	−0.605	0.081	0.592	Spot Fail
45	−0.094	−0.605	0.086	0.592	Spot Fail
61	−0.094	−0.605	0.086	0.592	Spot Fail
76	−0.094	−0.605	0.086	0.592	Spot Fail
88	−0.091	−0.605	0.083	0.592	Spot Fail
97	−0.094	−0.602	0.086	0.592	Spot Fail
90	−0.615	−0.605	2.350	0.592	Pass
91	−0.625	−0.605	2.410	0.592	Pass
⋮	⋮	⋮	⋮	⋮	⋮

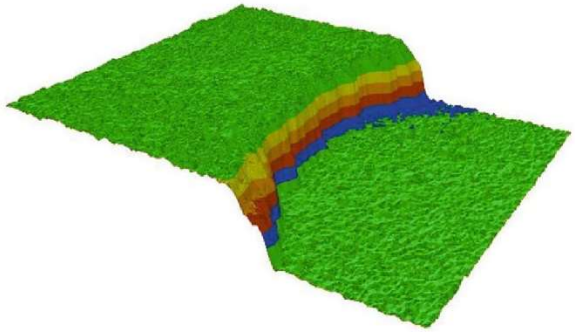
An example curve test result [Source: Integra Technologies]

3.4 Four Dimensional Scanning Electron Microscopy:

The stereo-photogrammetry technique requires computationally intensive image processing, but the presence of fast computers in recent years has resulted in more interest in this method [20]. A general stereo photogrammetric technique consists of three stages,

i. Acquisition stage: Acquires images of the same area of the specimen at 2 or more perspectives
ii. Matching stage: Matches the acquired images to find points corresponding to the same position

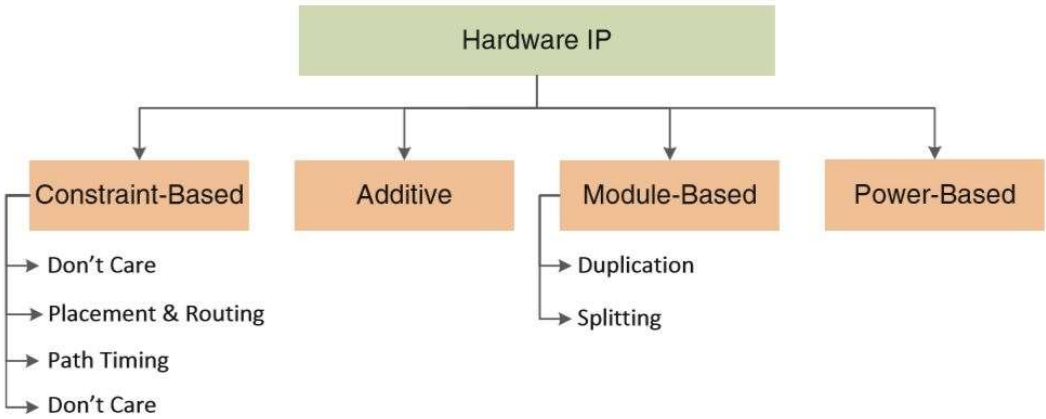
iii. Depth extraction stage: Extracts the third dimension based on projection geometry to obtain the 3D model



Four-dimensional image of a dimple, with colors represent sample material

3.5 WATERMARKING

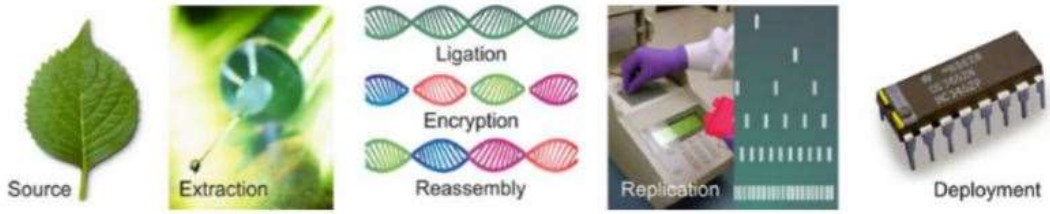
Watermarking is not a new concept and has been widely adopted in the field of digital media. Images, audio, and videos contain watermarks that prove the authorship of the media. Digital media watermarks can be visible or invisible; the visible ones are logos/signatures that appear on the media itself and invisible watermarks are those that have been incorporated into the media item and are invisible due to imperfections in the human audio-visual system.



IP watermarking strategies

3.6 DNA MARKING

Deoxyribonucleic acid (DNA) provides a unique identification where botanical DNA taggants are applied on the package. This essentially tags the electronic components to trace them throughout the component supply chain. Plant DNA is scrambled to create new and unique

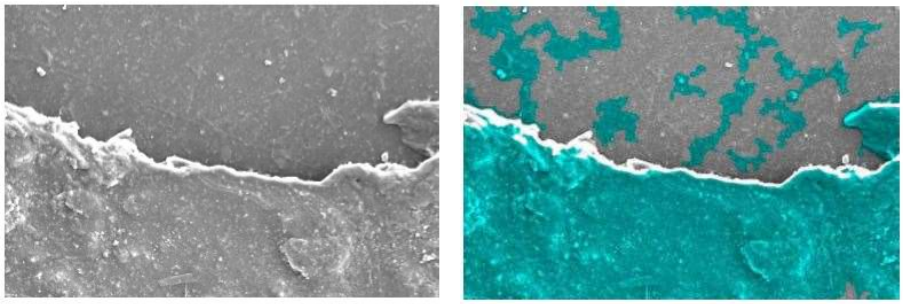


The creation and deployment of DNA taggants

genetic sequences, and these sequences of DNA are integrated with inks. These inks are then applied on the packages of the IC at the end of the packaging process. DNA marking provides unique protection against copying of the ID created by the DNA taggants. Unlike other marking techniques, the marks created by the DNA cannot be simulated, copied and/or reproduced.

3.7 BASIC IMAGE PROCESSING APPROACH

There are very few methods (2 to 4) has been invented to detect Counterfeit ICs using Image Processing Techniques. Previously we have developed Texture based segmentation and Detection which recently has been published at IEEE 2017

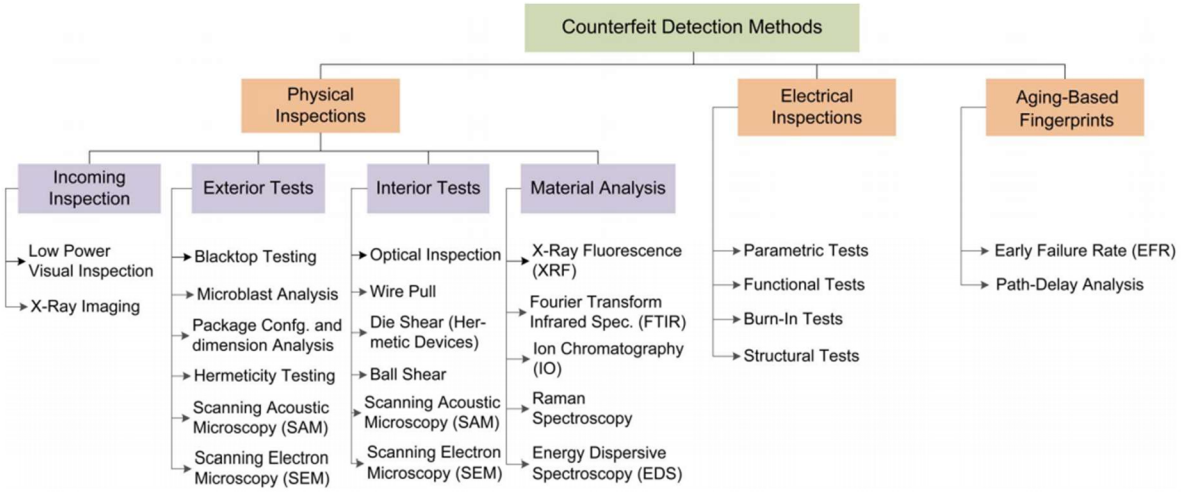


Texture based image before and after

Currently we are focusing on the counterfeit IC pin detection.

4. PROBLEM DESCRIPTION

Although there are lots of methods to detect counterfeit devices, most of them are destructive, and none can cover 100% of counterfeit types. Moreover, the detection methods have to evolve because the counterfeiting techniques may adapt to them. Furthermore, we need large numbers of detection techniques to deal with the large number of counterfeiting possibilities. Over the past several years, a specialized service of testing has been created for detecting counterfeit components. The figure below shows a generalized taxonomy of counterfeit detection methods.



Although counterfeiting seems like a grave problem, very little has been done in regard to its prevention. There are a few reasons as to why it has been proved difficult to carry on a mass scale reform against counterfeiting.

First, many counterfeits often go undetected and are installed into the finished product because they are indistinguishable from the original ones in aspects of functionality although differ a lot in quality. Even if a counterfeit is detected, it is not possible for the inspector to detect all the subtleties of the defective component. Compounding the problem, Original Component Manufacturers are often unwilling to aid in the identification of suspect parts purchased outside of their approved distributors. They want to sell current products or products through approved sources and do not want to encourage the use of unauthorized vendors.

Second, there is not one central clearinghouse for this information. Typically, when a counterfeit is detected, companies typically just refuse to pay for them and discard them. Organizations like ERAI compile counterfeit information but the sources are only their member companies. Thus chances are that a huge number of counterfeit products go undetected.

Third, there is a stigma associated with possessing counterfeits. Initially, companies reported counterfeit discoveries in their inspection but a mere association with counterfeit products was hugely discredited by the media tarnishing their reputation in the market. Even if the company may have better success in counterfeit prevention than their competition, reporting such discovery hurts their sales thus generating a fear of reporting such components. And if the crime is not reported, there is little that can be done to prevent it.

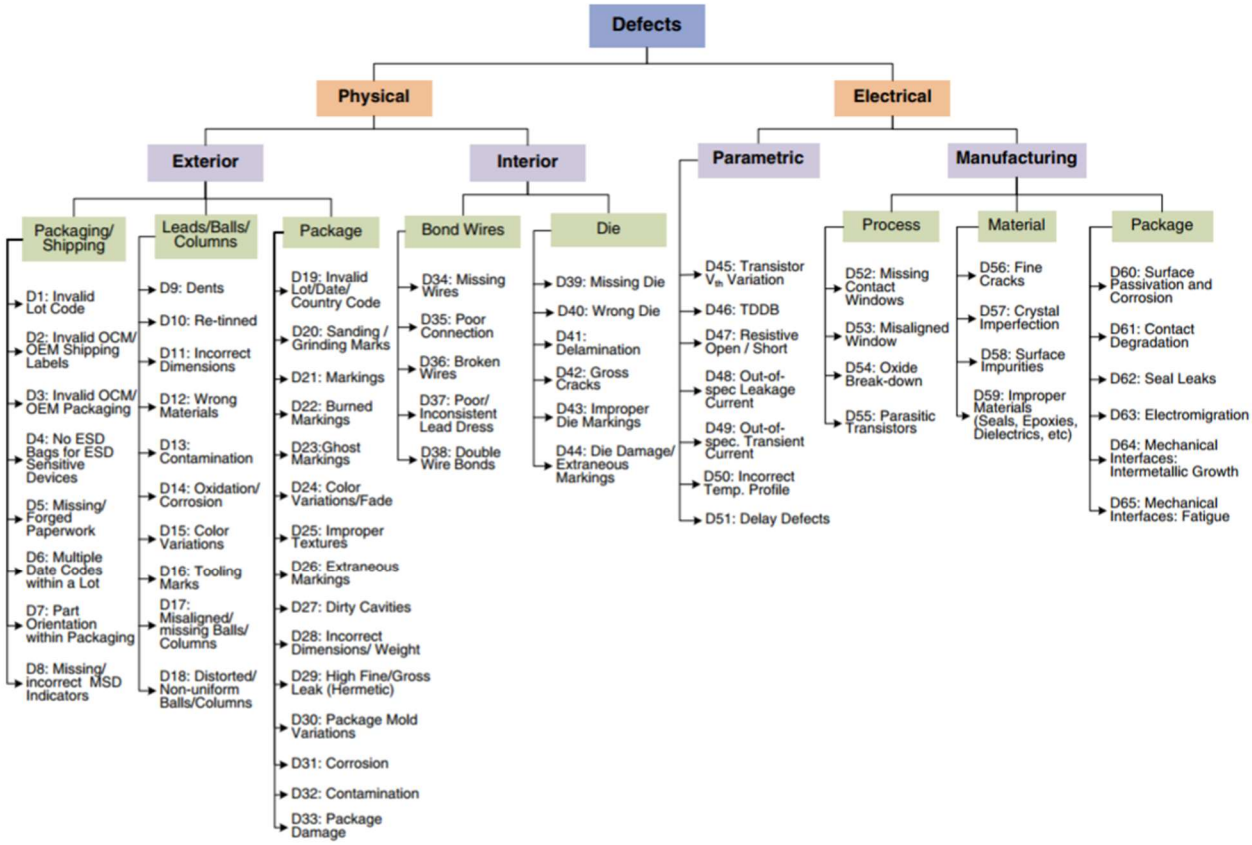
Fourth, the law enforcement and government agencies involved in counterfeit prevention have limited resources. There are numerous organizations that have agents and individuals investigating and developing plans to deal with counterfeit electronic components; the FBI, ICE, IRS, Defense Criminal Investigative Service (DCIS), Naval Criminal Investigative Service (NCIS), DOD, NASA, Government Accountability Office (GAO) and many others are all aware of the problem. However, in regard to the main investigative agencies, the FBI and ICE, the electronic community does not lobby for action as the industries do.

So, instead of focusing on prevention, the companies within the electronics industry currently, individually, focus on finding and eliminating counterfeits on a case-by-case basis. So there are large number of challenges ahead as well as future research scopes.

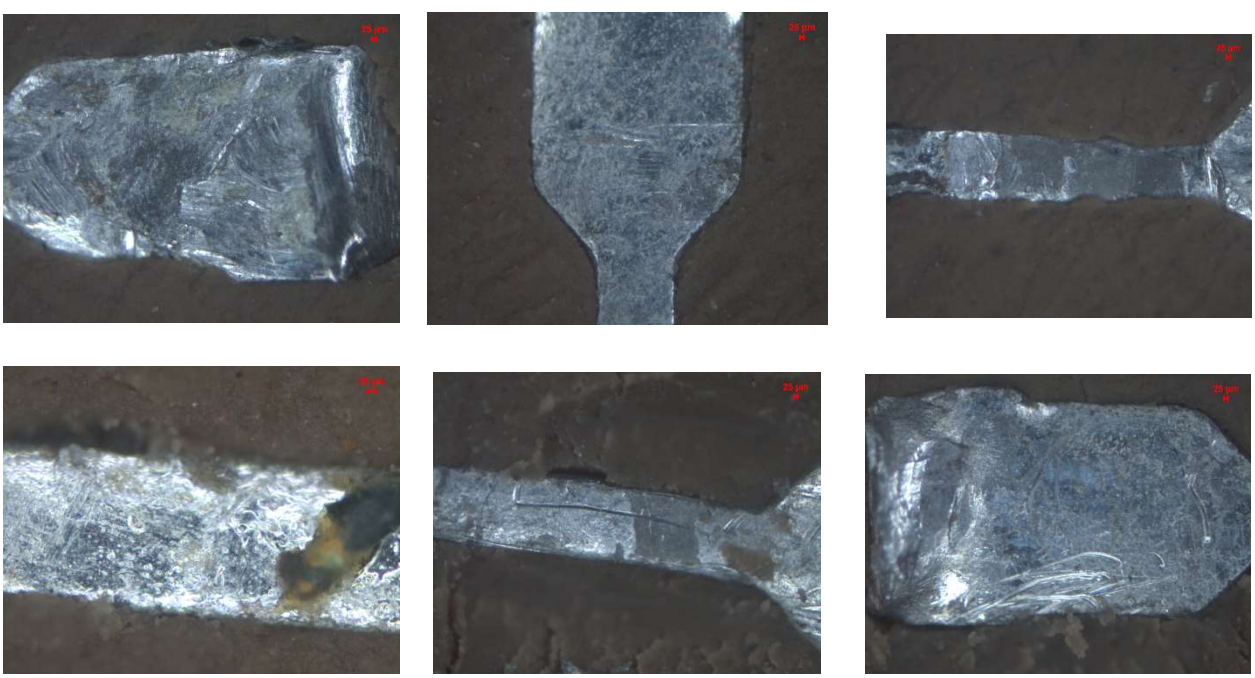
In this paper, my approach for detecting counterfeit IC will be based on several image processing techniques. Image analysis on the shape of an IC will help to detect if the IC is counterfeit or not. But before approaching the problem and techniques, I note down some salient features of counterfeit ICs. They are as follows:

1. The pins of the ICs have irregular width and shape.
2. The groove on the surface denoting the pin numbers have irregular position, even if they bear the same company logo. The grooves may also be absent in case of some ICs.
3. The pin surface textures may vary a lot among the same batch of ICs. Some may even have rust like textures along with smooth surface texture.
4. The company logo or the IC number may be overwritten using color and inscribed with a new name or number.
5. The pin surface may have a lot of scratches instead of the smooth surface texture which is expected.
6. Sometimes, ICs have distinct textures or colors on the body surface or on the pins.
7. The pins of the ICs, sometimes, have a slight inclination instead of being straight as expected, which is hard to detect with naked eyes.
8. Sometimes the new batch of revised and modified versions of ICs as recently manufactured and distributed by a company may include newly manufactured old discarded versions of the same ICs.

The following figure shows the various range of defects associated with counterfeit electronic components



Below is given the zoomed image of defective pins of ICs which is captured using an optical microscope:



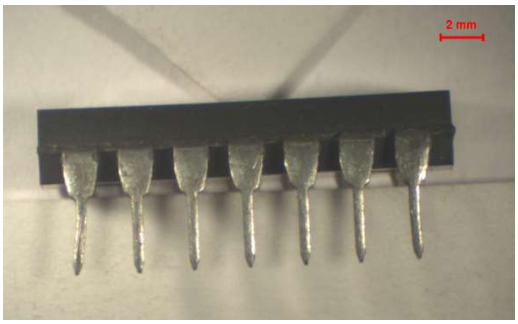
5. APPROACH

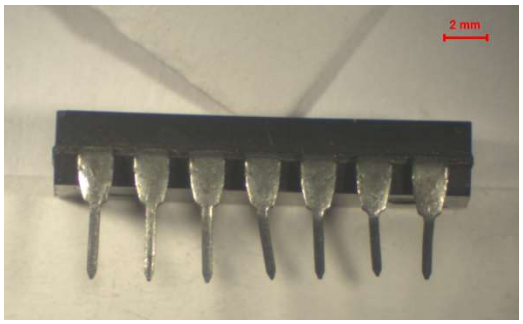
Though there are several methods and techniques to detect counterfeit ICs and to avoid them, all of them are really costly and takes time. So from the beginning, our focus was to detect if an IC is counterfeit in a small time and to make a system which will be cheap enough so that anybody from any lab or industries can use it easily. This paper focuses on image shape analysis processes to detect counterfeit. The following flow chart shows a diagram of the counterfeit IC detection method we implement. The main advantage of the proposed technique lies in its relatively low computational complexity, ease of automation (since no sample preparation is required), and its ability to achieve high classification accuracy by employing relatively low-end equipment and widely available software.

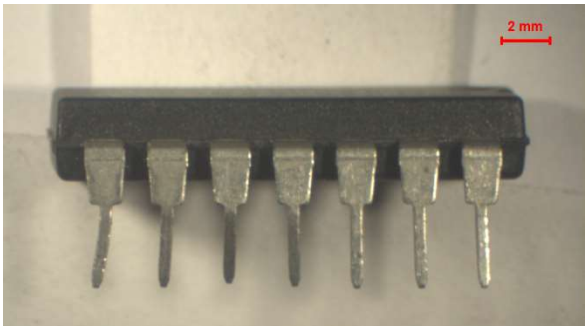
5.1 Image Acquisition using Optical Microscopy:

The optical microscope, often referred to as light microscope, is a type of microscope which uses visible light and a system of lenses to magnify images of small samples. For this project, an Optical microscope was used for image acquisition in our experiments.
The large field of view detector was chosen to be able to capture the entire die in a single image. In order to simulate real situations and further assess the robustness of our method, samples were mounted with different orientations and/or were tilted by the sample holder. We preferred to capture the images with a light background as the contrast of the body and pins with respect to a light background would make it easier for object detection and extraction in the future steps.

Below is shown a few examples of acquired images from optical microscope:







5.2 Image Registration and Feature Extraction:

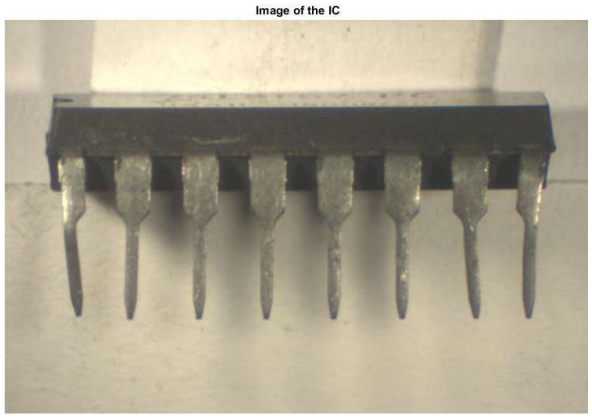
A number of mathematical approaches were explored for registration and feature extraction of the IC images obtained by microscopy. The basic approach was object detection and recognition to detect the pins of the IC from the images obtained from optical microscope. For object detection we used feature detection using Hough transform, edge detection using Sobel and Canny methods and Point feature matching all of which was implemented in MATLAB.

Point Feature Matching:
Mutual information based registration was initially considered to register the IC optical microscope images. Mutual information is a measure of the dependence among variables. When two gray scale images are correctly aligned, the dependence between their gray scale values is maximized, and consequently the mutual information value between the two images is also maximized. While this registration algorithm was successful in registering optical microscope images captured under the same operating parameters, this method did not perform in a satisfactory manner when attempting to register images to a base image taken under different optical operating parameters. In addition this registration algorithm was slow to converge for larger images. This algorithm can detect objects despite a scale change or in-plane rotation. It is also robust to small amount of out-of-plane rotation and occlusion.

Step 1: Read Images

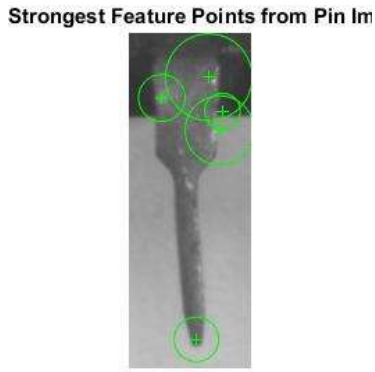
- Read the reference image containing the pin of interest.

- Read the target image containing the corresponding IC.

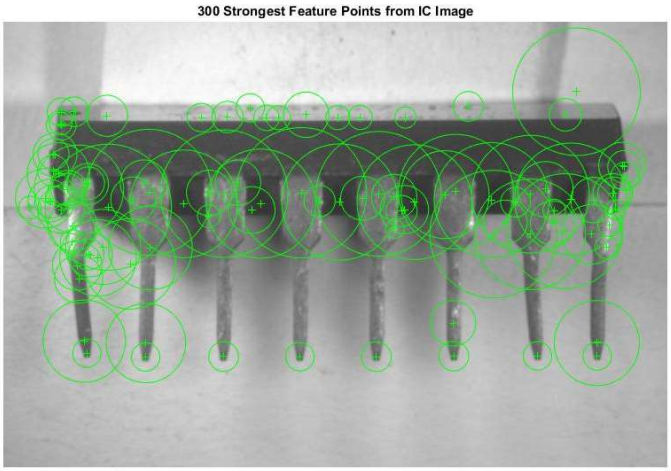


Step 2: Detect Feature Points

- Detect feature points in both images.
- Visualize the strongest feature points found in the reference image.



- Visualize the strongest feature points found in the target image.

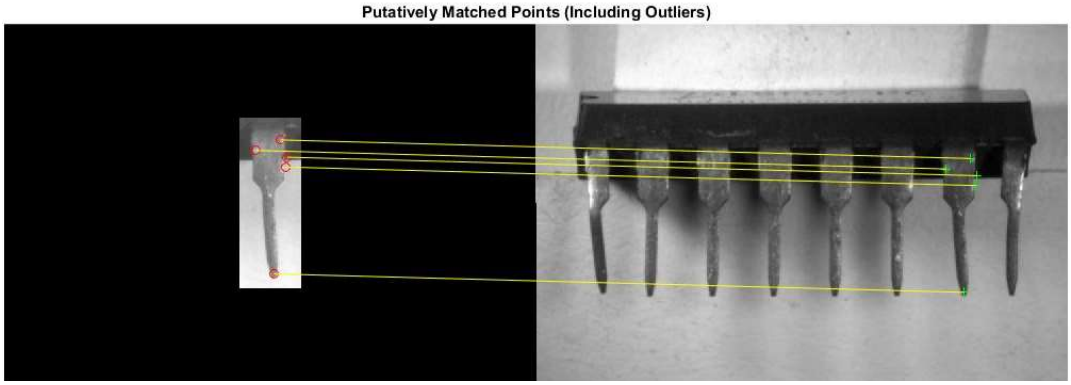


Step 3: Extract Feature Descriptors

Extract feature descriptors at the interest points in both images.

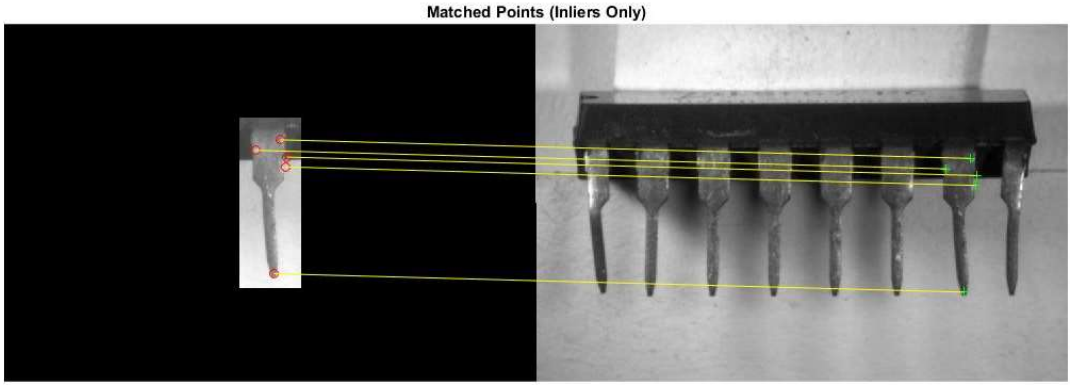
Step 4: Find Putative Point Matches

- Match the features using their descriptors.
- Display putatively matched features.

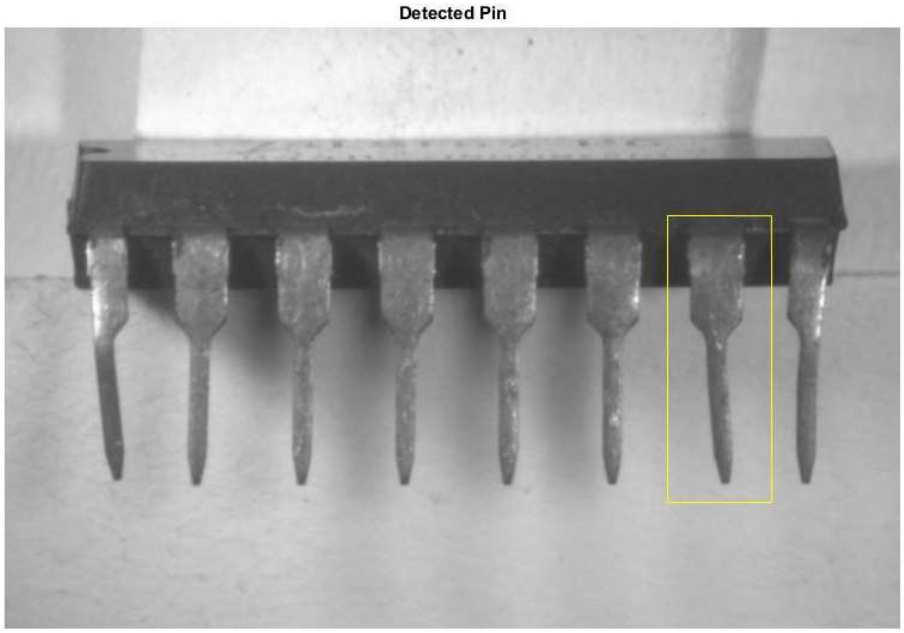


Step 5: Locate the Object in the Scene Using Putative Matches

- `estimateGeometricTransform` calculates the transformation relating the matched points, while eliminating outliers. This transformation allows us to localize the object in the scene.
- Display the matching point pairs with the outliers removed



- Get the bounding polygon of the reference image.
- Transform the polygon into the coordinate system of the target image. The transformed polygon indicates the location of the object in the scene.
- Display the detected object.

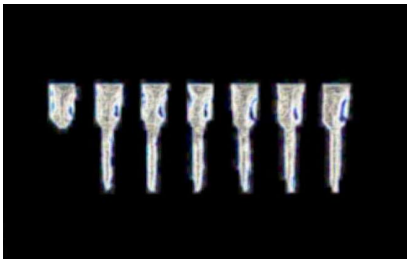


5.3 Mask Based Segmentation:

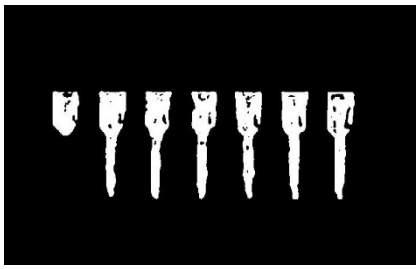
Segmentation by defining masks was explored as a viable option. The RGB images obtained from the microscope was at first converted to black and white, and the regions were demarcated using a certain threshold value. Image segmentation was applied to obtain masks from the image which was essentially restricted to the pins. Now, this extracted mask was used to segment further images. Although, it gave satisfactory results for certain images, images with different orientation and level of zoom caused irregular results and hence this method reached its limitation.

1. Read the mask image and convert it to a grayscale image.

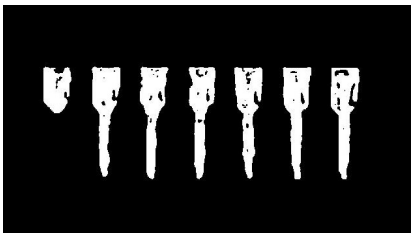
2. Apply entropy filtering on it so that the background becomes dull and the pins are bright.



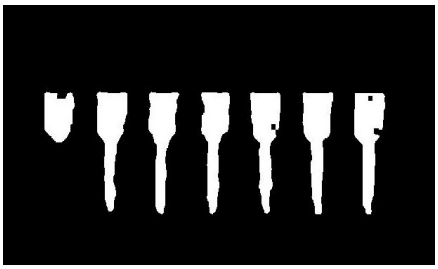
3. The entropy filter output is converted to binary image with a threshold value of 0.6



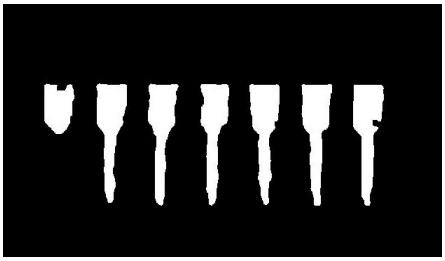
4. The smaller connected portions from the binary mask image is removed using the `bwareaopen` function, thus removing connected objects less than 5000 pixels.



5. The cleaned binary image is taken and morphological closing operation is done on it using a 9x9 'true' matrix structuring element which closes further gaps in the mask.

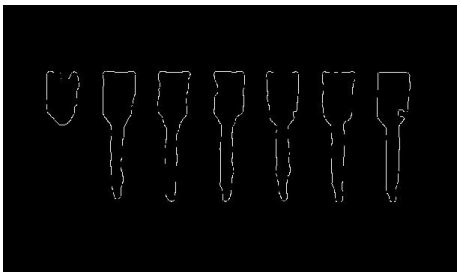


6. On the morphologically closed image, we apply `imfill` to close the holes.



7. The output of step 6 is the final mask which will be used for segmentation.

8. Edge detection is done on the final mask using Canny method for future processing.



9. The binary mask developed through the above steps is taken and is used in segmentation of the original image using the MATLAB 'Image Segmenter' app.

The problem was to apply such mask to extract pins from the IC images even with dissimilar orientation or sizes. So, the option of image recognition using deep learning and convolutional neural networks was explored and the work in this approach is still in its infant stage as further trials are being carried on and other suitable methods are being considered.

5.4 Predicted Hypothesis for IC Classification:

We use two different strategies to classify our data. Our first approach is based on a support vector machine (SVM) with a radial basis function kernel. The other classification approach is based on a deep neural network.

5.4.1 Support Vector Machine (SVM) Method:

SVM works by creating a hyperplane in a high dimensional space that can act as a discriminator between two classes. SVM achieves this separation by minimizing the following function:

$$f(w,\xi) = \frac{1}{2}w^T w + C\sum_{i=1}^{N}\xi_i \qquad (1)$$

Given the following constraints:

$$y_i(w^T\phi(x_i) + b) \geq 1 - \xi_i \qquad \text{for } i = 1, \dots, N \qquad (2)$$
$$\xi_i \geq 0 \qquad \text{for } i = 1, \dots, N \qquad (3)$$

where in Eq.(1), N is the total number of training samples, w represents a weight vector, C is the reciprocal of the regularization parameter used to avoid over fitting when the classification problem contains non-separable or noisy data points, and ξ_i is a slack variable for the ith training sample. In Eq.(2), y_i represents the class label (1 or-1), b represents a bias term, x_i represents the data vector corresponding to the ith training sample and $\varphi(.)$ is a kernel function that maps the input vector to a higher dimensional space. We used the radial basis function kernel

$$\phi(x, x_i) = \exp\left(-\frac{1}{2\sigma^2} * \|x - x_i\|^2\right) \qquad (4)$$

Where σ is the width parameter and x_i is the ith training sample. The SVM is set up with two classification parameters, as shown in Eqs. (1) and (4). The particular values of the SVM parameters, C and σ, were obtained through an exhaustive grid search. From the entire set of authentic and counterfeit ICs we randomly generated 4 different training and test partitions, each containing the same number of authentic and counterfeit samples. The training part of each partition was used to train the SVM with the specific C and σ values. The test part of each partition was used to determine the accuracy of the SVM. After the accuracy of SVM for each of the four test sets was computed, the average accuracy of the four sets was calculated and used as the qualifying metric to determine the best σ and C parameters.

5.4.2 Classification using Deep Neural Net:

The deep neural network consists of one input layer followed by a stacked auto encoder and a multilayer neural network classifier. The overall architecture results in four hidden layers and one output layer, as shown in figure below.

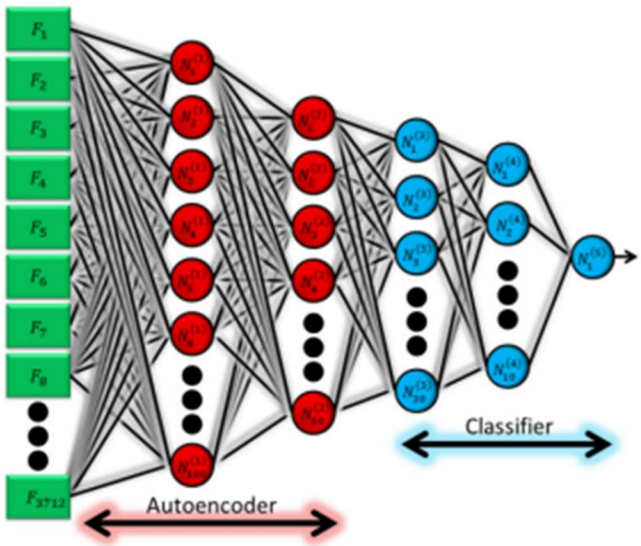


Fig: Deep neural network used in our classification approach. The squares labeled F represent the input to the network. The neurons labeled N_i^j in red represent the ith neuron in the jth layer of the stacked auto encoder. The neurons labeled N_i^j in blue represent the ith neuron in the jth layer in the classification part of the network.

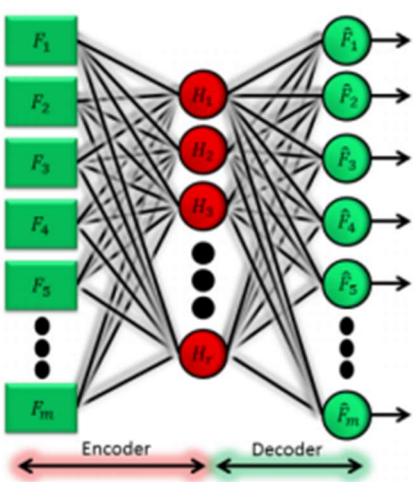


Fig: Encoder-Decoder network setup. The neurons $H_1, \ldots H_r$ represent the current layer L being trained. $F_1, \ldots F_m$ represents the inputs from the previous layer $L - 1$ and $\hat{F}_1, \ldots, \hat{F}_m$ represent the output neurons of the network which approximate a reconstruction of the input that is discarded after training.

A stacked auto encoder network makes up the input layer and first two hidden layers of the deep neural network. The third and fourth hidden layers, and the output layer of the neural network act as a classifier. The classifier part of the network and the auto encoder layers are trained differently to avoid the problems associated with training all the layers of a deep network using only backpropagation. The stacked auto encoder was trained using a greedy layer-wise approach as shown in Figure below.

In this training method, each layer is trained separately. For each layer L, the input to this layer from the previous layer L-1 is set as the target output. Layer L is then trained using stochastic gradient descent backpropagation for a given number of iterations, or until an error threshold is reached. After the weights of layer L have been trained, the process is repeated for the next layer L+1. The pseudocode for this training algorithm is given in figure below.

function **GreedyLayerTraining** $(x^1, \ldots, x^S)$

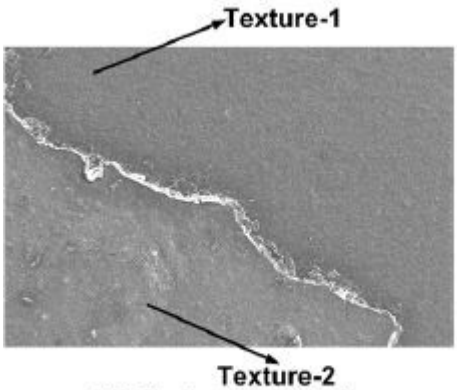
x_j^i = *The output vector from the* j^{th} *layer of the network for the* i^{th} *sample*
S = *The total number of samples*
$f_j(x)$ = *The function representation of the* j^{th} *layer of the network*
Q = *The total number of autoencoder layers in the network*
N_p = *The number of neurons in the* p^{th} *layer*

For L = 1: Q
    For $i = 1:S$
        For j = 1: L
            $x_{j-1}^i = f_{j-1}(x_{j-2}^i)$
        End
    1. Create a three layer autoencoder network K, with an input layer with N_{L-1} inputs, one hidden layer with N_L neurons and one output layer with N_{L-1} neurons.
    2. Train autoencoder network K with backpropgation using x_{j-1} as the input and x_{j-1} as the output for all S.
    3. After completion of training take the weights from the first hidden layer of K and apply them to the deep neural network layer L.
    End
End

After the auto encoder part of the network was set up using greedy layer-wise training, the entire network was trained using stochastic gradient descent backpropagation as part of the fine tuning step to adjust the weights of the entire deep neural network. For our classification part of the network we used two hidden layers, which were obtained through empirical testing. It is possible to approximate any continuous function with a single hidden layer comprised of a finite number of neurons according to the universal approximation theorem. However in our implementation, we chose two layers to reduce the overall complexity that a single hidden layer with a large number of neurons would create, and to improve the classification run time.
The auto-encoder is trained using a large number of counterfeit IC image data such that it acts as a classifier which takes an image as an input and as an output classifies it into original IC or as counterfeit IC.

5.5 Image Texture Analysis:

An IC package is essentially a plastic component whose surface is textured after removing it from the mold. Counterfeiters usually try to achieve similar and uniform texture over the entire package surface to hide previous markings. This texture difference is not visible to the naked eye, but with the help of a microscope one can get many signs that may help to identify the fake surface. Fig. below shows one such case where two different textures are visible on a single IC package surface.



(d) Texture mismatch
(on same IC surface)

We used two methods for texture analysis and segmentation-

5.5.1 Laws' Texture Energy Method: Analysis of each image produces a 9-dimensional feature vector at each pixel position. Now, the golden IC textural feature values were compared with those obtained from the test-case images. If the values were closer to the values of the golden IC then it is a probable original IC. More the difference, more is its probability to be a counterfeit IC.

5.5.2 LBP based Texture Metric Comparison: For each images, including the golden IC image, one LBP generated histogram is obtained. Next step is to compare the test image histogram with the golden IC image histogram, using the Euclidean-distance measure and KL-distance measure. In our scheme, the 256 possible histogram values were clustered in ten different bins consisting of values 0 - 25, 26 - 49, …. 226 - 255, i.e. n = 10 in Eqn. (1) and Eqn. (2)

Euclidean Distance between the two histograms: (Eqn 1)

$$L_1(H_1, H_2) = \frac{1}{N}\sqrt{\sum_{i=1}^{n}(H_1[i] - H_2[i])^2}$$

Where *n* represents the total number of bins of the histograms and $N = \sum_{i=1}^{n} H_1[i] = \sum_{i=1}^{n} H_2$

Kullback-Leibler Divergence Measure (KLDivergence or KL-Distance): (Eqn 2)

KL divergence is a measure of the non-symmetric difference between two discrete distributions $\{p_i\}$ and $\{q_i\}$, defined as:

$$D_{KL} = \sum_{i=1}^{n} p_i \log_2 \left(\frac{p_i}{q_i} \right)$$

Where *n* is the total number of points in the two distributions. These probabilities can be calculated by normalizing the bin sizes with respect to the total sample size.

6. EXPERIMENTAL RESULTS

6.1 Classification Results:

A deep neural network can outperform an SVM due to their difference in classifier architecture. SVMs are considered a type 2 shallow architecture, and it has been proved that a classifier with a deep architecture, such as a deep neural network, can outperform a shallow architecture. Examining the parameters used to create the SVM, we can also see certain architectural problems relating to our data set. In a normal classification approach, it is undesirable to closely fit a function to the data. SVM avoids this problem by using a slack variable, which is controlled by the parameter C. Use of this parameter allows some data points to fall inside the region of separation or on the wrong side of the hyperplane, which in our particular case results in more misclassifications. Alternatively, neural networks have no parameter to control over-fitting. Because the data points in the circled region are so close together, it is necessary to fit the function precisely to the data points to achieve high accuracy.

Comparing the SVM to the deep neural network, we can see there is a discrepancy in performance between the two classifiers. This difference can be attributed to two main factors, the distribution of the data set created by linear PCA and the stacked auto-encoder, and the difference in classifier architecture. It has been proved that auto-encoders are more effective in reducing the dimensionality of a data set than standard linear PCA. In our data set, this same trend is also apparent and we are still now developing our neural network design so that it can work faster than other soft wares which can detect counterfeit ICs. We are also trying to develop a different hypothesis to detect embedded counterfeit circuits of an IC. The confusion matrix of our deep neural network and confusion matrix for SVM are

Deep neural network confusion matrix

	Predicted Positive	*Predicted Negative*
Positive Class	True Positive(TP)=7	False Negative(FN)=0
Negative Class	False Positive(FP)=0	True Negative(TN)=9

Support vector machine confusion matrix

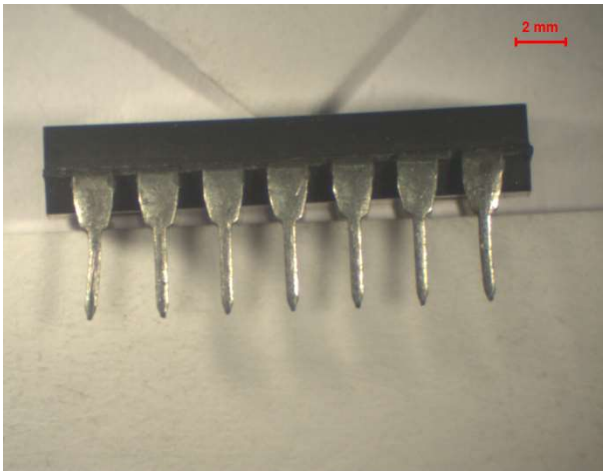
	Predicted Positive	*Predicted Negative*
Positive Class	True Positive(TP)=5	False Negative(FN)=2
Negative Class	False Positive(FP)=0	True Negative(TN)=9

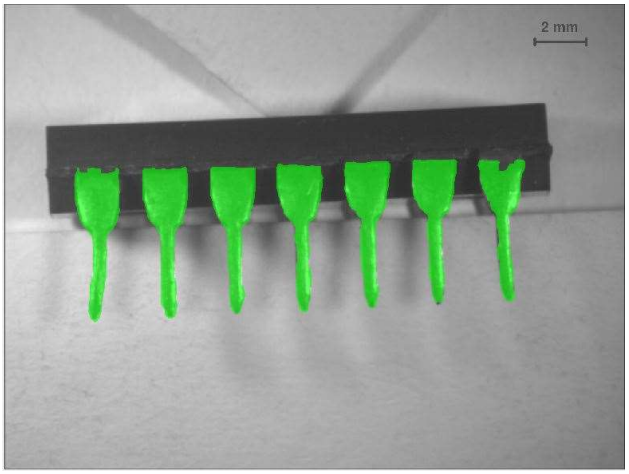
Due to absence of availability and guarantee we assumed some ICs as golden and trained our machine using them and used other ICs as test data. We expect that in our next step we will be able to show proper golden ICs using our machine without proper training data.

6.2 Mask Based Segmentation Results:

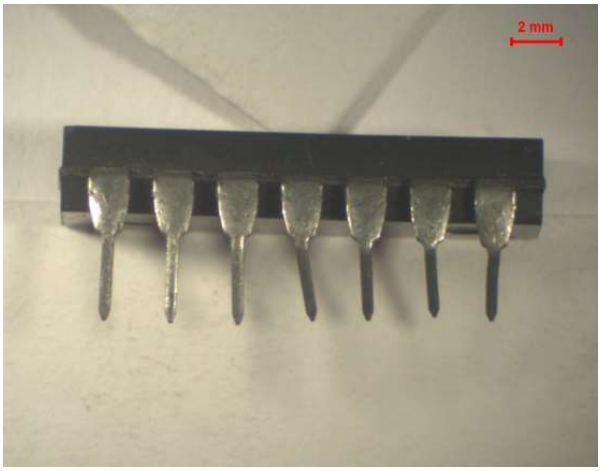
6.2.1 Some of our successful results of before after differences are below-

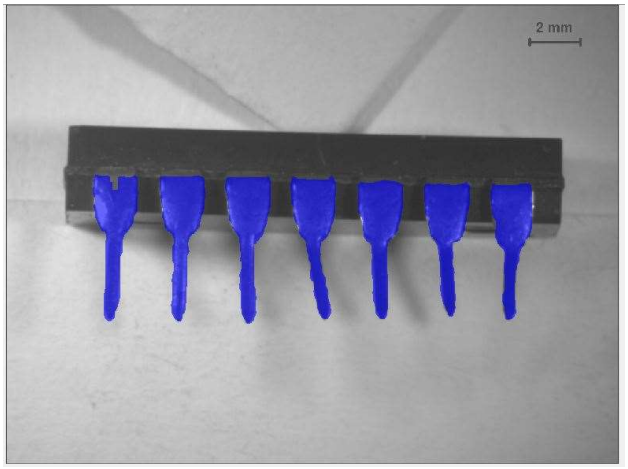
a. **Result-1:**



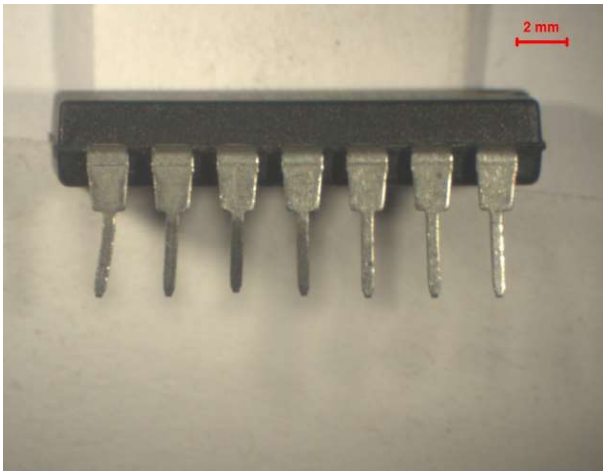


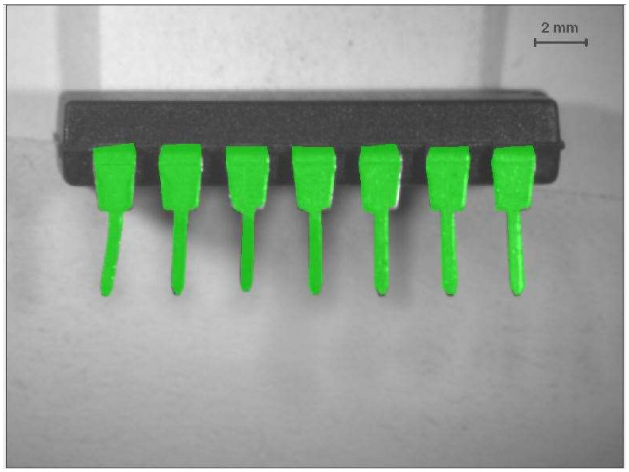
b. **Result-2:**



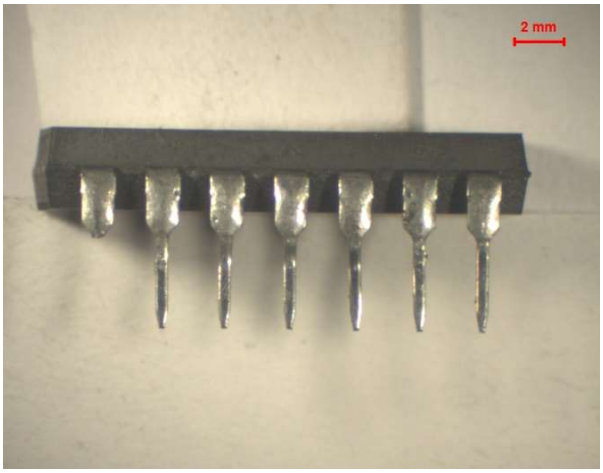


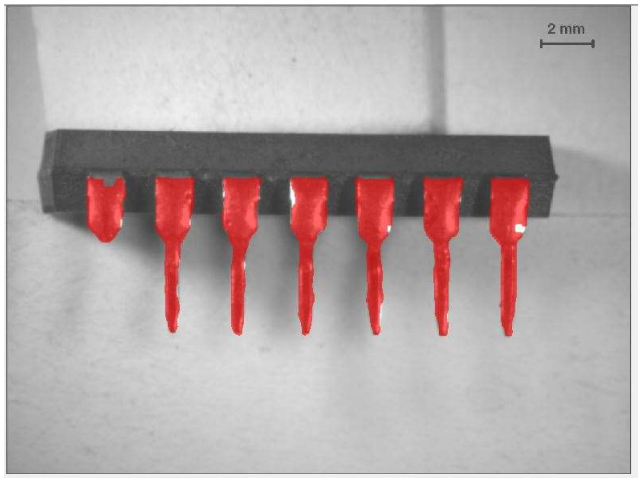
c. **Result-3:**



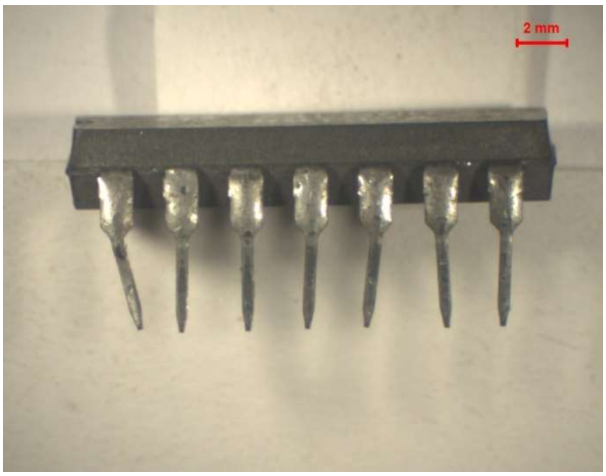


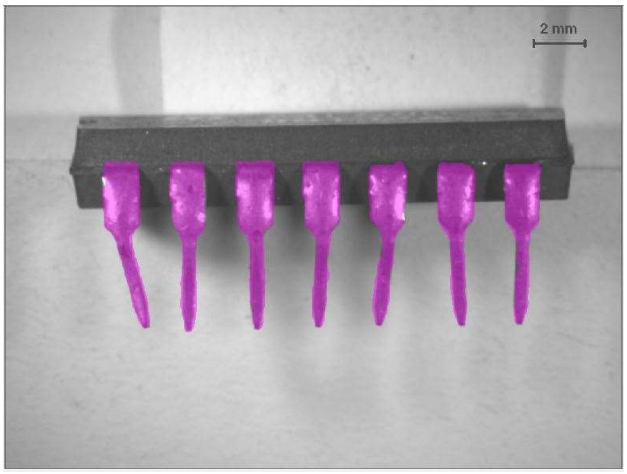
d. **Result-4:**





e. **Result-5:**

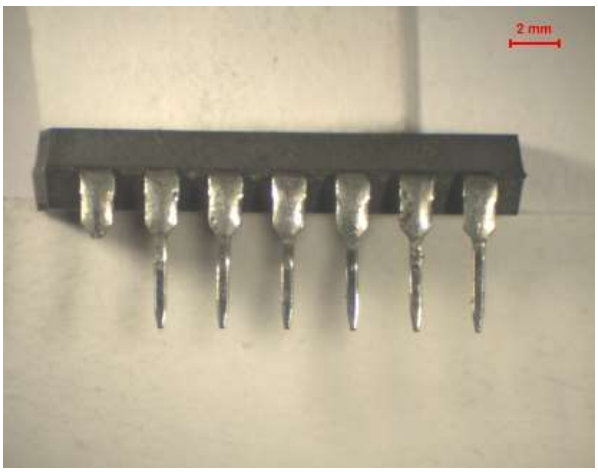


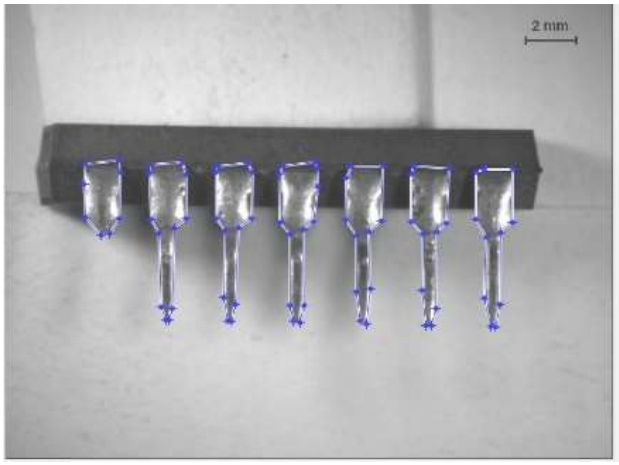


6.2.2 Results showing detection of IC pins:

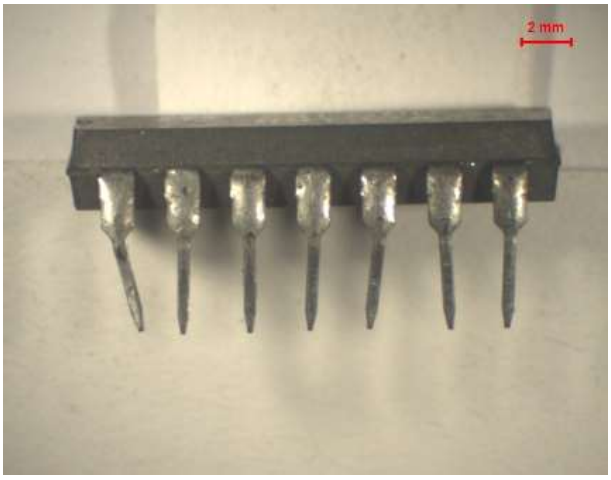
Detection of the IC pin was the initial task and then we colored the pins so that we can understand which part to segment clearly. Some of our detection result samples are shown as below.

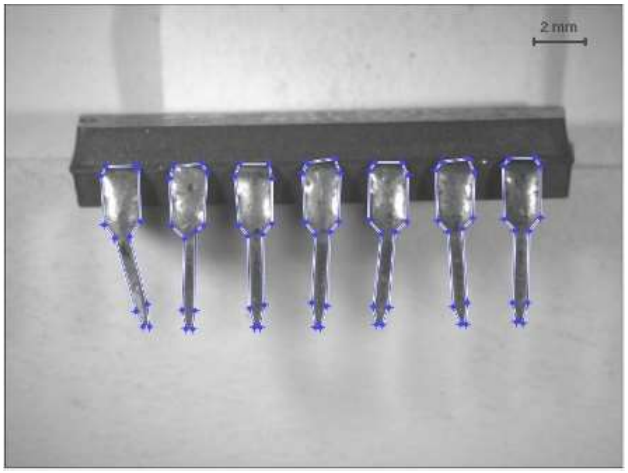
a. Result-1:





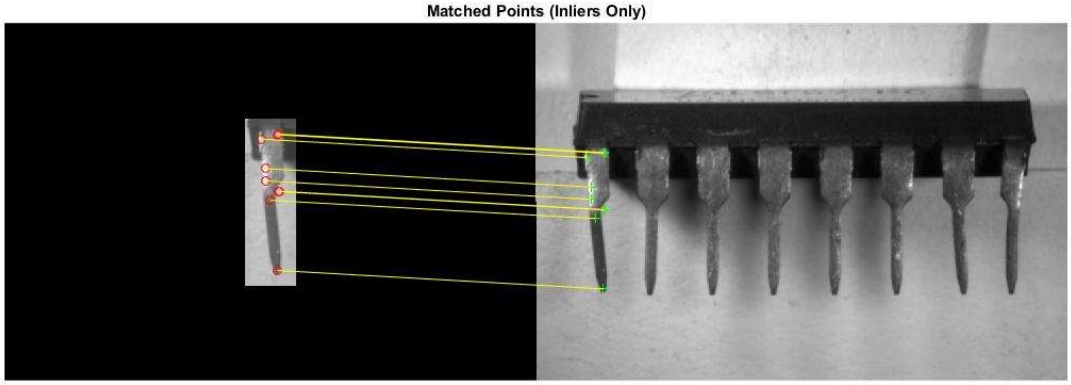
b. Result-2:

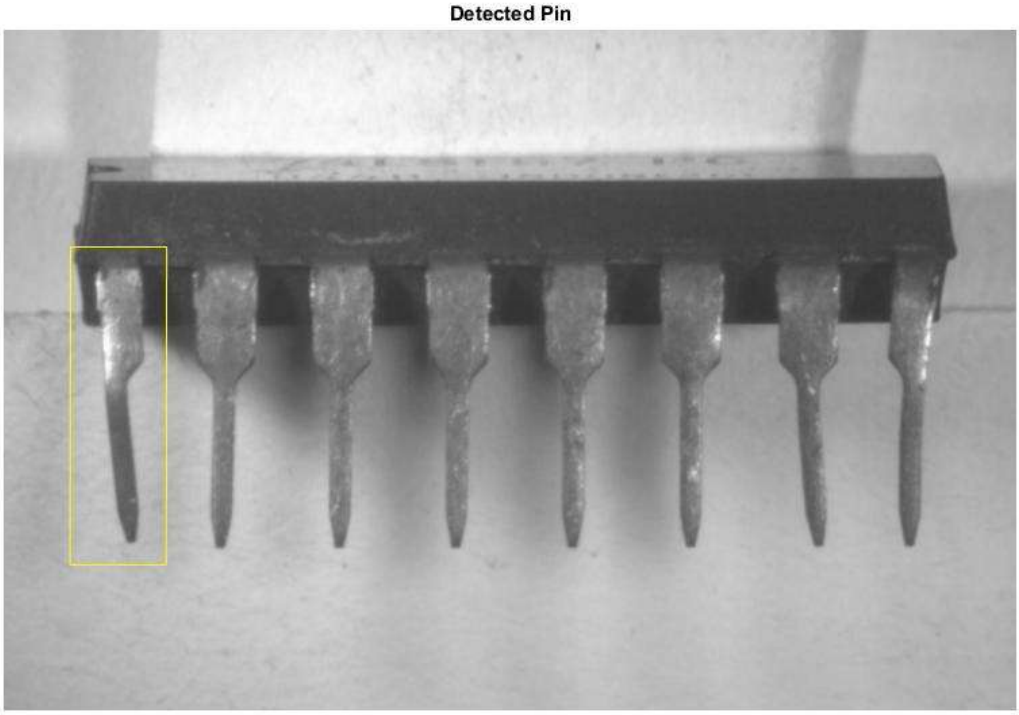




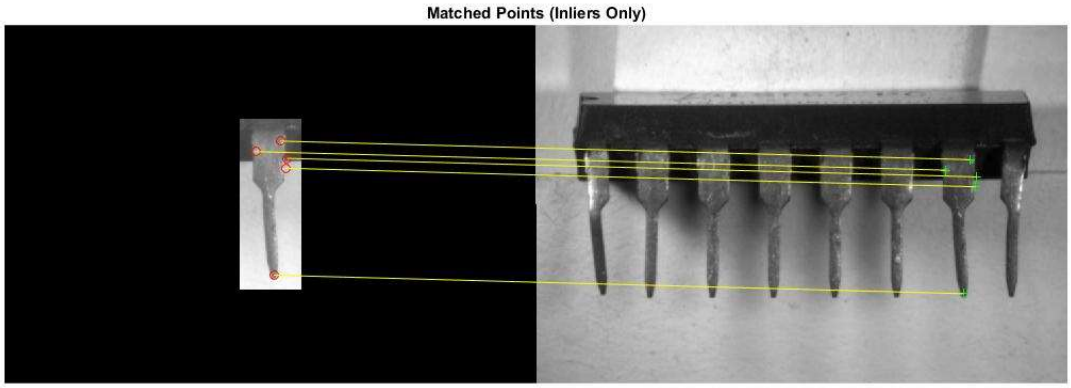
6.3 Point Feature Detection Results:

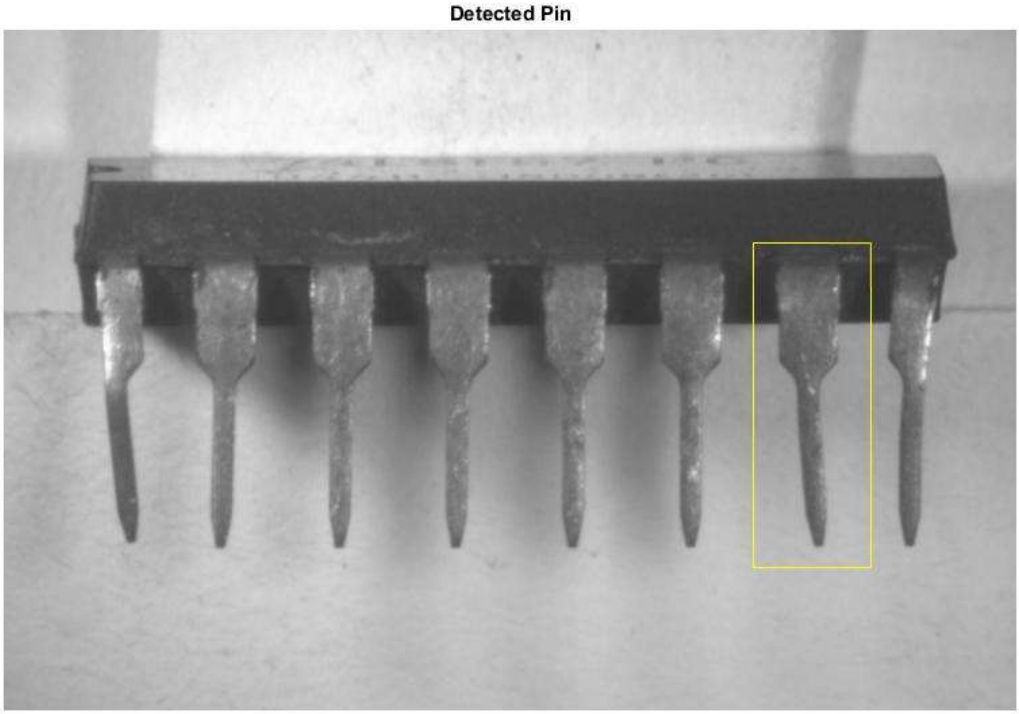
Result-1:





Result-2:

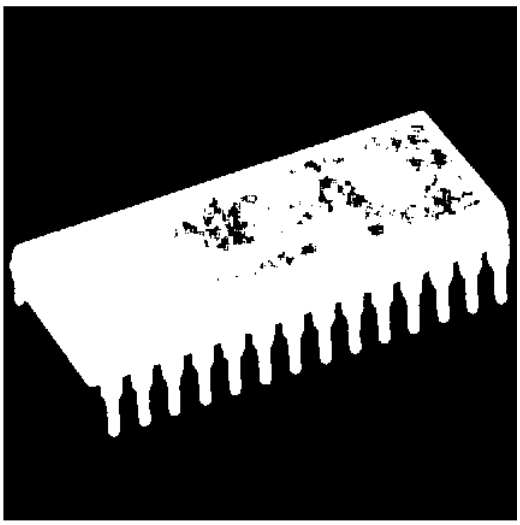


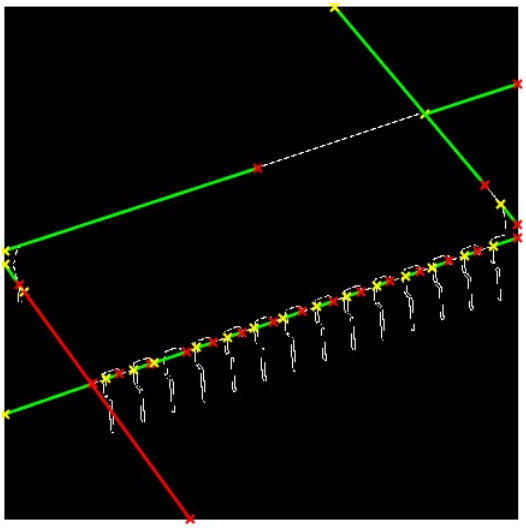


6.4 Challenges Faced:

- The result of our previous process using mask based segmentation was not so satisfactory and was facing some computational complexity problem. We tried to segment pins from any IC using one mask but we failed because of angular capturing problem so we looked after deep learning methods.
- Before attending the mask based segmentation problem we also tried Gabor filter, texture based segmentation, Hough Transform and UV ray techniques but their algorithms became so long that we started searching for other techniques. Though these processes supposed to work but it required more effort and time as they are not so simple like machine learning and deep learning techniques.
- Some unsuccessful results that we faced while researching are shown below

6.5 Unsuccessful Results:





7. CONCLUSION

The method was successful in the detection of pins but the database of masks needed to be prepared separately for the individual images. Hence it was time consuming as well as intensive from the aspect of manual effort. Hence, we have to look for better efficient methods which would ease the detection, would be unsupervised, less intensive and preferably more time and computationally efficient.

8. FUTURE WORK

Our next work will be to develop deep learning algorithm to extract features from an IC image and analyze their properties. We will build an app to filter original ICs from a bunch of IC images using advanced image processing techniques. We will try to build our app supported in Linux, Windows, Mac and Android operating systems.

9. REFERENCES

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